

**Birla Institute of Applied Sciences, Bhimtal**  
**Artificial Intelligence (CST-022)**  
**Assignment-2**

**Submission Date- 26-04-2026**

1. Define a Knowledge-Based Agent (KBA). Explain the role of the Inference Engine and the Knowledge Base (KB).
2. Differentiate between Weak Slot-Filler and Strong Slot-Filler structures with examples for each.
3. Differentiate between forward chaining and backward chaining.
4. Discuss the major challenges in Knowledge Representation, specifically the Frame Problem, Granularity, and Representing Defaults.
5. Explain Propositional Logic. What are its syntax and semantics? What are the limitations of Propositional Logic for knowledge representation? Convert the following English sentences into Propositional Logic:
  - a) **If it is raining, and then the ground is wet.**
  - b) **It is raining and it is cold.**
  - c) **Either it is sunny or it is cloudy.**
6. Convert the following English sentences into First-Order Logic (FOL):
  - a) **All humans are mortal.**
  - b) **There exists a student who likes AI.**
  - c) **For every x, if x is a cat, then x is an animal.**
7. Find the Most General Unifier (MGU) for the following pairs of literals, or state why they cannot be unified:
  - **$f(x, g(y))$  and  $f(A, g(B))$**
  - **$P(x, x)$  and  $P(y, f(y))$**
  - **$Loves(John, x)$  and  $Loves(y, Father(y))$**
8. Consider the following sentences to prove the Resolution.
  - **John likes all kinds of food**
  - **Apples are food**
  - **Chicken is food**
  - **Anything anyone eats and isn't killed by is food**
  - **Bill eats peanuts and is still alive**
  - **Sue eats everything bill eats**
  - i) Translate these sentences into formulas in predicate logic
  - ii) Prove that John likes peanuts using backward chaining
  - iii) Convert the formulas of a part into clause form
  - iv) Prove that John likes peanuts using resolution